

Valuing the Benefits of Health and Safety Regulation¹

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A goal of many public regulatory programs is the protection or enhancement of human health. If such programs are to be subjected to close analytical scrutiny, if not full-blown cost-benefit analysis, it is important to have correct measures of their benefits. This is no simple task. Not only is it often difficult to map regulatory policy into physical improvements in health or safety (less sickness, fewer accidents, etc.), but also it is difficult to translate changes in health status into dollar valuations. This is perhaps most difficult when the policy outputs are "lives saved," or—more correctly—lives prolonged, but difficulties arise even when health benefits take the form of reductions in acute or chronic illness unrelated to premature mortality.

Perhaps the most common means of valuing reduced morbidity is the cost of illness (COI) approach, whereby benefits are valued by savings in direct, out-of-pocket expenses arising from illness or injury (e.g., medicines and doctor and hospital charges), as well as opportunity costs incurred, foregone earnings being the most obvious example (Cooper and Rice [2]; Lave and Seskin [9]).² One problem with the COI approach is that it ignores any disutility arising from illness or injury. Perhaps more importantly, the COI approach generally does not take into account the defensive or averting expenditures that individuals can and do make to protect their well-being (Mishan [10]; Fisher and Zeckhauser [5]; and, more recently, Courant and Porter [3]; Shibata and Winrich [12]; Harford [6]). This omission can be critical. If defensive or averting measures fully protect

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²As is often the case with ad hoc constructs like the cost of illness, not everyone uses the same definition. Most studies of particular illness incidence, including pollution studies, count only medical care and lost wages. However, attempts to estimate total health care costs sometimes include what we have termed "defensive expenditures" as a cost of illness. See Hodgson and Meiners [7] for a discussion.

against insults to safety and health, a regulation-induced reduction in such insults might not lead to any observed health improvement, but would be beneficial nonetheless by alleviating the need for such protective measures.

Recently, Courant and Porter [3] and Harford [6] have looked at changes in averting expenditures alone (in a nonhealth framework) as a possible measure of willingness to pay for the alleviation of insults. They conclude that averting expenditures may under- or overstate true benefits. This raises an important question for practical health benefit estimation: If data were available for both the direct and the indirect costs of illness or injury as well as the averting measures, what would be the relationship between their sum and true willingness to pay? This is the question that we address below.

We first present a simple static model of constrained utility maximization in which illness or injury not only reduces the time which can be devoted to labor or leisure and induces remedial medical expenditures, but also causes disutility. In this model, individuals can make defensive or averting expenditures to lessen their chances of illness or injury, thus influencing directly their health status. Next, we recast the problem in terms of the indirect utility function, derive the conceptually correct measure of the benefits of reductions in the health threats that individuals face, and compare this measure with the cost of illness plus the change in defensive expenditures. True benefits are found to exceed this sum for all practical cases. In Section 3 we find that this conclusion holds under a number of extensions to the simple model.

1. THE BASIC MODEL

Consider an individual with a utility function of the form

$$U = U(X, L, S) \quad (1)$$

where X represents expenditures on all non-health-related goods,³ L is leisure time per period, and S is time spent either ill or injured per period ($U_X, U_L > 0$; $U_S < 0$; $U_{XX}, U_{LL}, U_{SS} < 0$). Time spent sick depends on some broadly defined environmental threat which could take the form of air pollution, exposure to a harmful substance in the workplace or in a foodstuff or other consumer product, or other hazard.⁴ For simplicity we refer to all such threats as pollution, P .

Sickness also depends on preventive defensive or averting expenditures, D , which can reduce the likelihood of any illness or injury or shorten the

³We find it more convenient to make utility a function of expenditures on various classes of goods rather than a composite good, as is customary.

⁴Although the disutility of sickness presumably depends on its intensity as well as its duration, we assume that all illnesses are equally intense.

duration of those that may occur. For instance, if P is interpreted as air pollution, D might take the form of automobile or home air conditioning or air filtration. If P is a drinking water contaminant, D might represent the purchase of bottled water or a filtration system for the tap. If P represents the threat of accident posed by a particularly dangerous stretch of road, D might even take the form of more cautious driving with its attendant time costs.

Thus,

$$S = S(D, P) \quad (2)$$

where $S_D < 0 < S_P$, and $S_{DD}, S_{PP} > 0$. Medical expenses, M , follow any illness or injury and are assumed to be equal to some daily cost, m , times the number of days spent ill or injured. In other words,

$$M = mS(D, P). \quad (3)$$

Finally, the individual is assumed to receive per-period nonwage income I (from capital or transfer payments) and to work for wage rate w .

The individual's decision problem can then be expressed as

$$\begin{aligned} \max_{X, L, D} \mathcal{L} = & U(X, L, S) + \lambda [I + wT - wL - wS(D, P) \\ & - X - D - mS(D, P)] + \mu [T - L - S(D, P)] \quad (4) \end{aligned}$$

where T is total time available per period. The bracketed expression multiplying λ is the "full income" constraint as introduced by Becker [1]. The second constraint ensures that the sum of leisure time and time spent ill does not exceed the total time available. The first-order conditions are

$$\mathcal{L}_X = U_X - \lambda = 0 \quad (5)$$

$$\mathcal{L}_L = U_L - w\lambda - \mu = 0 \quad (6)$$

$$\mathcal{L}_D = U_S S_D - w\lambda S_D - \lambda - \lambda m S_D - \mu S_D = 0 \quad (7)$$

$$\mathcal{L}_\lambda = I + wT - wL - wS(D, P) - X - D - mS = 0 \quad (8)$$

$$\mathcal{L}_\mu = T - L - S \geq 0; \mu(T - L - S) = 0. \quad (9)$$

For the balance of this paper, we assume that all individuals work some positive number of hours per period. Thus, from (9), $\mu = 0$. It follows, then, that (5) and (6) represent the tradeoff between labor and leisure (X is total expenditures on other goods, so that its price is unity).

Equation (7) can be rewritten to shed light on defensive expenditures by individuals. Specifically,

$$\frac{U_S S_D}{\lambda} - wS_D - mS_D = 1. \quad (10)$$

Thus, at the margin, a dollar's worth of defensive expenditure is comprised of three parts: the dollar value of disutility arising from additional sick time ($U_S S_D / \lambda$), the opportunity cost of illness valued at the wage ($-wS_D$), and the out-of-pocket expenses caused by increased illness ($-mS_D$).

To value the health benefits arising from a reduction in P , we employ the indirect utility function (see Courant and Porter [3]; Just *et al.* [8]). The optimal values for the choice variables X , L , and D depend on w , I , and P . The indirect utility function, V , gives the maximum utility obtainable for a given set of parameter values:

$$\begin{aligned} V &= V(I, P, w) \\ &= U(X, L, S(D, P)) \\ &\quad + \lambda [I + wT - wL - wS(D, P) - X - D - mS(D, P)] \\ &\quad + \mu [T - L - S(D, P)] \end{aligned} \quad (11)$$

where X , L , D , λ , and μ are functions of the parameters I , P , and w .

Suppose that I and w are fixed. If P changes, say from P_0 to P_1 , utility changes from V_0 to V_1 , where

$$V_0 = V(I, P_0, w) \quad \text{and} \quad V_1 = V(I, P_1, w). \quad (12)$$

If $P_0 < P_1$, then $V_0 > V_1$. For fixed wages, the only way to keep an individual indifferent to a change in pollution is to change I . This is illustrated in Fig. 1a. When P increases from P_0 to P_1 with no change in income, the individual drops from the indifference curve $V = V_0$ to $V = V_1$. To leave the individual indifferent to the change in P , I must be increased from I_0 to I_1 . We define the benefit to the individual of the change in air quality as the measure $I_1 - I_0$.⁵

As suggested by Fig. 1, holding V constant while P varies defines I implicitly as a function of P , which we denote as $I^*(P)$. Thus, $V(I^*(P), P) = V_0$.⁶ The total derivative of V with respect to P is set equal to zero along the indifference curve $V = V_0$. Then

$$\frac{dV}{dP} = V_I \frac{dI^*(P)}{dP} + V_P = 0 \quad \text{or} \quad \frac{dI^*(P)}{dP} = -\frac{V_P}{V_I}. \quad (13)$$

⁵ Note that it is not always possible to keep the individual on the same indifference curve. In the case illustrated in Fig. 1b, there is no change in income that compensates the individual for an increase in pollution from P_0 to P_1 . This possibility can arise only for increases in pollution. The change in income associated with a decrease in pollution is always defined. In any event, we are interested primarily in small changes for which this possibility is unlikely to arise; hence, it is ignored in what follows.

⁶ Definition of I as a function of P requires $V_I \neq 0$, but we assume, as is customary, that $V_I > 0$.